Dec-ZoomSIB-Relay without a server: proofs

Lower bound, regret, communication, robustness and agreement

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Abstract

We analyse the serverless version of Fed-ZoomSIB, where N agents communicate only along the edges of a graph $\mathcal{G}$. Every message is flooded, so it reaches every agent within Λ rounds; on a static connected graph Λ is the diameter D. (1) No serverless algorithm beats $\Omega((NT)^{2/3})$ (Theorem 5). On a ring or a path it must also pay $\Omega(N \min(T, D)^{1/3})$, which dominates for short horizons (Theorem 7). (2) With high probability, Dec-ZoomSIB-Relay has regret equal to the server bound plus an additive delay term that does not grow with the horizon: $\tilde{O}\big((dNT)^{2/3} + N\Lambda K\big)$ for synchronised epochs (Theorems 17 and 22), and $\tilde{O}\big((dNT)^{2/3} + KF'\big)$, with F' the farness, for the published-only relay, a one-line variant of the relay we run (Theorem 27). The relay bound holds for every trigger threshold γ, with a factor $\sqrt{1+N\gamma}$ on the UCB term only, so it also covers the constant γ we run, at the flood relay's communication. (3) It sends $O\big(|E|\, N\, N_{\mathrm{b}}^2 \log_{1+\gamma}(NT)\big)$ scalars, logarithmic in T for a fixed grid (Theorem 29), with $O(N_{\mathrm{b}}+N)$ memory per agent (Proposition 30). (4) Directed graphs, link delays and time-varying graphs only change Λ (Corollary 32). (5) If agents freeze different directions, the regret grows by at most $4L_{f'}X_{\max}\,\eta\,NT$ (Theorem 33). Any server stopping rule can be run without a server, with exact agreement, at a cost of Λ rounds (Proposition 34). (6) Phase 1 can be made differentially private with one noisy message per agent, at a cost of a factor $1 + O(\sqrt{dN}/\varepsilon)$ in the direction error (Theorem 35). The analysis uses *synchronised epochs*: all agents act on the same delayed pooled table plus their own recent pulls. This keeps the confidence argument of the server case unchanged and removes the $\sqrt{N}$ loss of the earlier sketch.

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1 Setting, notation and imported results

Bandit model. There are N agents and T rounds. At round t agent i observes K contexts $X_{i,t} = (x_{i,t,1}, \dots, x_{i,t,K})$, drawn i.i.d. from $\mathcal{D}$ on $\mathbb{R}^d$, independently across agents and rounds. It pulls an arm $a_{i,t}$ and observes

$$y_{i,t} = f(\langle x_{i,t,a_{i,t}}, \theta^* \rangle) + \eta_{i,t}.$$

Both θ^* (with $\|\theta^*\|_1 = 1$) and f are unknown and shared by all agents. Write $z = \langle x, \theta^* \rangle$ for the *true index* of an arm x. The network regret is

$$R_T(N) = \sum_{i=1}^{N} \sum_{t=1}^{T} r_{i,t}, \qquad r_{i,t} = \max_{a \le K} f(\langle x_{i,t,a}, \theta^* \rangle) - f(\langle x_{i,t,a_{i,t}}, \theta^* \rangle).$$

Order of pulls. Within a round, all N agents act in parallel. For the analysis we order all pulls *lexicographically* by (round, agent): $(t, i) \prec (t', i')$ iff $t < t'$, or $t = t'$ and $i < i'$. Let $\mathcal{F}_{t,i}$ be the σ-field generated by all contexts, actions and rewards strictly before (t, i) in this order, together with $X_{i,t}$ and the algorithm's internal randomness.

Assumption 1 (DBG Assumptions 1–3)**.** (i) The noise variables $\eta_{i,t}$ are i.i.d., independent of all contexts and of the algorithm's internal randomness, with $\mathbb{E}[\eta] = 0$ and $\mathbb{E}[e^{\lambda\eta}] \le e^{\lambda^2\sigma^2/2}$ for all $\lambda \in \mathbb{R}$. In particular $\eta_{i,t}$ is independent of $(\mathcal{F}_{t,i}, a_{i,t})$.

(ii) $\mathbb{E}[S_j(X)^2] \le M$ for the score $S = -\nabla \log p$.

(iii) Each coordinate of X is κ-sub-Gaussian. On $|z| \le X_{\max}$ we have $|f| \le L_f$ and $|f'| \le L_{f'}$, where $X_{\max} = \kappa\sqrt{2\log(2dNTK/\delta)}$. With probability $\ge 1-\delta$, every one of the NTK contexts satisfies $\|x\|_\infty \le X_{\max}$. Call this event E_x.

(iv) $\mu^* = \mathbb{E}[f'(\langle X, \theta^* \rangle)] > 0$.

DBG state (i) only as conditional sub-Gaussianity. Their Stein bound (Lemma 2 below) is proved for i.i.d. pairs (x, y), which needs i.i.d. noise in Phase 1, so we state it explicitly. Their hard instance (Lemma 3) has i.i.d. $\mathcal{N}(0, 1)$ noise. Phase 2 of the epoch protocol (Sections 3–4) uses only the consequence $\mathbb{E}[e^{\lambda\eta_{i,t}} \mid \mathcal{F}_{t,i}, a_{i,t}] \le e^{\lambda^2\sigma^2/2}$. The published-only relay (Section 5) orders pulls by arrival time rather than by (round, agent), so it uses the full i.i.d. statement (i).

Communication model. The agents are the vertices of a graph $\mathcal{G} = (V, E)$ with $|V| = N$. Each round has two steps: a *decision step*, in which every agent pulls an arm, and then a *communication step*, in which every agent may send messages to its neighbours. A message sent in the communication step of round s is received before the decision step of round $s+1$. *Flooding* means that an agent forwards every message the first time it receives it, to all its neighbours, and never again. A message received in step s is forwarded in step $s+1$, so it moves one hop per step. (For the epoch protocol, faster relaying would only shorten the delivery times below. Section 5 uses exactly this one-hop-per-step flooding, because its proof needs the exact arrival time of every record.)

Assumption 2 (Delivery time)**.** There is a known integer $\Lambda \geq 1$ such that a message that is flooded from any agent in the communication step of round s has reached every agent by the end of the communication step of round $s + \Lambda - 1$. Each agent knows N, Λ and the round number.

Lemma 1 (Flooding on a static graph)**.** *If $\mathcal{G}$ is static, connected and has diameter D, Assumption 2 holds with $\Lambda = D$.*

Proof. Let agent a flood a message in step s. We show by induction on $h \geq 1$ that every agent at distance h from a holds the message after step $s + h - 1$. For $h = 1$: neighbours of a receive it in step s. For the inductive step, let v be at distance $h+1$. It has a neighbour u at distance h, which holds the message after step $s + h - 1$. If u received it in step $s + h - 1$ or earlier, it forwards it no later than step $s + h$. So v holds it after step $s + h$. Every agent is at distance at most D. □

Imported results. The following results are proved elsewhere and used as black boxes. Here DBG refers to Dey, Bhore & Ghosh (2026).

Lemma 2 (Stein estimator; DBG Lemma B.1 with Lemma 2.1)**.** *Let $\hat{\theta} = \frac{1}{n}\sum_{s=1}^{n} \varphi_\tau(y_s S(x_s))$ be the truncated Stein estimator on n i.i.d. pairs, with $\tau = \sqrt{3(\sigma^2 + L_f^2)Mn/\log(2d/\delta)}$, and let $\hat{\theta}_0 = \hat{\theta}/\|\hat{\theta}\|_1$. If $C_\theta d\sqrt{\log(2d/\delta)/n} \leq \mu^*/2$, then with probability $\geq 1 - \delta$,*

$$\|\hat{\theta}_0 - \theta^*\|_1 \leq \varepsilon_0(n) := \frac{4C_\theta d}{\mu^*}\sqrt{\frac{\log(2d/\delta)}{n}}, \qquad C_\theta = \tfrac{4\sqrt{3}}{3} + 2\sqrt{2}\sqrt{M(\sigma^2 + L_f^2)}.$$

Lemma 3 (Single-agent lower bound; DBG Theorem 5.1)**.** *There are $c > 0$ and n_0 such that for every horizon $n \geq n_0$ and every single-agent policy π, $\sup \mathbb{E}[R_n(\pi)] \geq c\, n^{2/3}$. The supremum is over the following instances, which satisfy Assumption 1 with $L_f = 1$, $L_{f'} = 3$ and $\sigma = \kappa = M = 1$: $d = 1$, $\theta^* = 1$, $\mathcal{D} = \mathcal{N}(0, 1)$, i.i.d. $\mathcal{N}(0, 1)$ noise, a link with $|f| \leq 1$ on all of $\mathbb{R}$, $\mu^* \geq c_1/(2 \log n)$, and $K = \lceil 4\lceil n^{1/3}\rceil \log n/\phi(1)\rceil$ arms per round. Here ϕ is the standard normal density and $c_1 > 0$ is a constant (DBG Lemma F.1).*

Lemma 4 (Martingale concentration; Azuma–Hoeffding for sub-Gaussian increments)**.** *Let $(\xi_m)_{m\geq 1}$ be adapted to a filtration $(\mathcal{H}_m)$ and satisfy $\mathbb{E}[e^{\lambda \xi_m} \mid \mathcal{H}_{m-1}] \leq e^{\lambda^2\sigma^2/2}$ for every $\lambda \in \mathbb{R}$. Then for every fixed $p \geq 1$ and $\delta' \in (0, 1)$,*

$$\mathbb{P}\Big(\Big|\sum_{m=1}^{p} \xi_m\Big| > \sigma\sqrt{2p\log(2/\delta')}\Big) \leq \delta'.$$

2 Lower bound

Theorem 5 (No serverless algorithm beats $(NT)^{2/3}$)**.** *Let $\mathcal{A}$ be any N-agent algorithm, on any graph and with any communication. This includes the extreme case where every agent sees every*

other agent's data after every round. Then for $NT \geq n_0$,

$$\sup_{\text{instances}} \mathbb{E}\big[R_T(N;\mathcal{A})\big] \;\geq\; c\,(NT)^{2/3},$$

where the supremum is over the instances of Lemma 3 with $n = NT$.

Proof. The proof is a reduction: from $\mathcal{A}$ we build a single-agent policy π with horizon $n = NT$ whose regret has the same law as the network regret of $\mathcal{A}$. Then we apply Lemma 3.

Step 1: relabelling time. Map the single-agent round $u \in \{1, \dots, NT\}$ to the pair

$$u \;\longleftrightarrow\; (t, i), \qquad u = (t-1)N + i, \quad t \in [T],\ i \in [N].$$

Step 2: the policy π. Policy π stores the complete state of $\mathcal{A}$: every agent, every message buffer, and every random seed. It draws the seeds itself. At its round $u = (t, i)$:

(a) it receives the context set X_u and gives it to the simulated agent i;
(b) the simulated agent i chooses a_u from its state after round $t-1$, using exactly the rule of $\mathcal{A}$;
(c) π plays a_u and stores the reward y_u without giving it to the simulation;
(d) if $i = N$, π gives the N stored rewards of round t to the simulated agents and runs one communication step of $\mathcal{A}$.

Each action a_u depends only on X_u, on rewards of rounds $u' \leq (t-1)N < u$, and on π's own randomness. So π is a valid single-agent policy.

Step 3: the laws coincide. Let H_t be the history (contexts, actions, rewards) of rounds $1, \dots, t$ of $\mathcal{A}$, and H'_t the history of rounds $1, \dots, tN$ of π, grouped in blocks of N. We show by induction on t that $H'_t \overset{d}{=} H_t$. The case $t = 0$ is trivial. Assume $H'_{t-1} \overset{d}{=} H_{t-1}$.

- In both systems the N context sets of round t are i.i.d. from $\mathcal{D}^K$ and independent of the past.
- Given the past and the contexts, the N actions are the same measurable function of these and of the seeds, in both systems. This is because π withholds the round-t rewards.
- Given the actions, each reward is $f(\langle x, \theta^* \rangle) + \eta$, with noise of the same conditional law in both systems.

So $H'_t \overset{d}{=} H_t$. The regret is the same function of the history in both systems, namely the sum of the NT per-pull gaps. Hence $R_{NT}(\pi) \overset{d}{=} R_T(N;\mathcal{A})$ on every instance, and

$$\sup \mathbb{E} R_T(N;\mathcal{A}) = \sup \mathbb{E} R_{NT}(\pi) \;\geq\; c\,(NT)^{2/3}$$

by Lemma 3. □

Remark 1 (The hard instances)**.** The hard instances depend on the horizon. Each agent sees $K \approx 4(NT)^{1/3}\log(NT)/\phi(1)$ arms per round, and μ^* is of order $1/\log(NT)$. DBG's minimax bound is stated in the same way. It is not a bound for a fixed K such as $K = 2$. On these instances the upper bound of Section 4 loses only logarithmic factors: K enters only through $X_{\max}$, and $1/\mu^*$ only through ε_0.

Remark 2. The proof uses only two facts: the agents are statistically identical (they share θ^*, f and $\mathcal{D}$), and within a round they act in parallel. It does not depend on $\mathcal{G}$. The next result uses the graph.

2.1 A lower bound that depends on the graph

Information moves one hop per round, so for a while each agent can only use data from its neighbourhood. Write $d(u,v)$ for the hop distance and $B(i,r) = \{u : d(u,i) \le r\}$. The *light cone* of agent i with radius R is the set of agent-rounds

$$C_i(R) = \{(u,t) : \ t \ge 1,\ d(u,i) + t \le R\}, \qquad n_i(R) = |C_i(R)| = \sum_{t=1}^{R} |B(i, R-t)|.$$

Lemma 6 (Light cones are closed)**.** *In any algorithm that communicates as in Section 1, the actions in $C_i(R)$ are functions of the algorithm's random seed and of the contexts and rewards in $C_i(R)$ only.*

Proof. We first show by induction on s that what agent u knows after the communication step of round s depends only on the seed and on the data of pulls (v, s') with $s' \le s$ and $d(u,v) \le s - s' + 1$. For $s = 1$, u knows its own pull ($d = 0$) and its neighbours' round-1 messages ($d = 1$). For the step, u adds its own round-s pull, and messages from neighbours w. These depend on w's round-s pull ($d(u,w) = 1$) and on what w knew after step $s-1$: by induction, pulls with $d(w,v) \le s - s'$, hence $d(u,v) \le s - s' + 1$.

The action of u at round t uses $X_{u,t}$ and what u knew after step $t-1$: pulls (v,s') with $s' < t$ and $d(u,v) \le t - s'$. If $(u,t) \in C_i(R)$, then $d(v,i) + s' \le d(u,i) + d(u,v) + s' \le d(u,i) + t \le R$, so $(v,s') \in C_i(R)$. □

Theorem 7 (Graph-dependent lower bound)**.** *Let $R \le T$, let S be a set of agents whose pairwise distances exceed $2R$, and let $n = \min_{i \in S} n_i(R)$. If n is large enough for Lemma 3, then every serverless algorithm satisfies*

$$\sup_{\text{instances}} \ \mathbb{E}\, R_T(N) \ \ge \ c' \, |S| \, n^{2/3}, \qquad c' = \tfrac{1}{256e},$$

where the supremum is over the hard instances of Lemma 3 for horizon n.

Proof. Idea. Each cone is a small, closed network. It behaves like one agent with n pulls, so it must pay $\Omega(n^{2/3})$. The cones are disjoint, so these costs add up. The only care needed is to make all cones pay under *one* common instance.

Cones are disjoint. If (u,t) were in $C_i(R) \cap C_{i'}(R)$, then $d(i,i') \le d(u,i) + d(u,i') \le 2R - 2t < 2R$.

Each cone is a single-agent policy. Order the pulls of $C_i(R)$ by (round, agent) and keep the first n. By Lemma 6, each of these pulls depends only on earlier rounds of the cone and its own contexts. As in the proof of Theorem 5, this is a single-agent policy π_i with horizon n that withholds same-round rewards. Its regret is at most the cone's share of $R_T(N)$, because regret is non-negative.

One instance for all cones. We follow DBG's proof of their Theorem 5.1 (their Appendix F), with $M = \lceil n^{1/3} \rceil$ bins, the null hypothesis $\beta^{(0)}$ and the spikes $\beta^{(j)}$. For cone i let N_j^i be the expected number of pulls of π_i in bin j under $\beta^{(0)}$, and A_j^i the event that π_i pulls bin j at least $n/2$ times.

- Since $\sum_j N_j^i \le n$, the set $J_i = \{j : N_j^i \le 2n/M\}$ has $|J_i| \ge M/2$.
- At most two of the events A_j^i can happen at once, so $\sum_j \mathbb{P}_{\beta^{(0)}}(A_j^i) \le 2$. Hence the set $B_i = \{j : \mathbb{P}_{\beta^{(0)}}(A_j^i) > 1/(4e)\}$ has $|B_i| < 8e$.

For $M \ge 32e$, $|J_i \setminus B_i| \ge M/4$ for every $i \in S$. Counting pairs, some bin j lies in $J_i \setminus B_i$ for at least $|S|/4$ agents i. Fix that j and such a cone i. DBG's Proposition F.1 gives $\mathrm{KL}(P^{\pi_i}_{\beta^{(0)}} \| P^{\pi_i}_{\beta^{(j)}}) \le 2\varepsilon^2 N_j^i \le 1$. The Bretagnolle–Huber inequality then gives $\mathbb{P}_{\beta^{(0)}}(A_j^i) + \mathbb{P}_{\beta^{(j)}}((A_j^i)^c) \ge 1/(2e)$, so $\mathbb{P}_{\beta^{(j)}}((A_j^i)^c) \ge 1/(4e)$. This is DBG's Case A, which gives $\mathbb{E}_{\beta^{(j)}} R(\pi_i) \ge n^{2/3}/(64e)$. Summing over the at least $|S|/4$ disjoint cones, $\mathbb{E}_{\beta^{(j)}} R_T(N) \ge |S|\, n^{2/3}/(256e)$. □

Corollary 8 (Rings and paths)**.** *On a ring or a path with N agents and diameter D, every serverless algorithm satisfies*

$$\sup \mathbb{E}\, R_T(N) \;\geq\; c'' \max\Big\{(NT)^{2/3},\; N\min(T,D)^{1/3}\Big\}$$

for an absolute constant $c'' > 0$, once $\min(T,D)$ is large enough for Lemma 3.

Proof. The first term is Theorem 5. For the second, take $R = \min(T, \lfloor (N-1)/4 \rfloor)$, which is of order $\min(T,D)$, and place the centres of S every $2R+1$ agents, so $|S| \geq \lfloor N/(2R+1) \rfloor \geq N/(6R)$. Balls of radius $r < R$ contain at least $r+1$ agents (at least $2r+1$ on a ring), so $n_i(R) \geq \sum_{r=0}^{R-1}(r+1) \geq R^2/2$. Theorem 7 gives $\Omega\big((N/R)\cdot R^{4/3}\big) = \Omega(NR^{1/3})$. □

Remark 3 (What this does and does not show)**.** The second term beats the first exactly when $N\min(T,D) \geq T^2$, for example on a ring when $T \lesssim \sqrt{ND}$. In that regime, a network on a ring *cannot* do as well as one agent with NT pulls: the graph really costs regret. For longer horizons this lower bound no longer sees the graph. The upper bounds pay an additive, horizon-free $O(KF')$ (Theorem 27), linear in the farness. Whether a term linear in the distances is necessary is open.

Remark 4 (Why the light-cone bound grows only like $D^{1/3}$)**.** A cone with n agent-rounds learns like one agent with n samples, and in a single-index bandit that already gives regret $n^{2/3}$, i.e. per-pull regret $n^{-1/3}$. The reason is smoothness: with $|f'| \leq L_{f'}$, a hidden good bin of width w can be better than its neighbours by at most $L_{f'}w$. Finding one of M such bins costs $\approx M/(L_{f'}w)^2$ samples, and DBG's choice $w = 1/M$ balances this at $n^{2/3}$. A linear-in-D lower bound would need hard instances in which far-away information is worth more than local exploration. We do not know such instances that satisfy Assumption 1 with fixed constants. Constructions of this "needle" type give only $n^{2/3}$ per cone.

3 The algorithm as analysed

The algorithm has a grid of bins on the index line and two phases. The table below fixes the notation.

Symbol	Meaning
T_0	exploration rounds per agent; Phase 2 starts at round T_1+1, $T_1 = T_0 + \Lambda$
$W = X_{\max}$, Δ	window and bin width; bins $I_1 = [-W, -W+\Delta]$ and $I_j = (-W+(j-1)\Delta,\, -W+j\Delta]$ for $2 \leq j \leq N_{\mathrm{b}} = \lceil 2W/\Delta \rceil$
$L = 2\Lambda$	slot length; slot boundaries $\beta_m = T_1 + mL$, $m \geq 0$, and $\beta_{-1} := \beta_0$
$\mathcal{E}$	set of epochs (a subset of slot boundaries); $\beta_0 \in \mathcal{E}$
$O_{k,j}(s)$	number of Phase-2 pulls by agent k into bin j in rounds $\leq s$
$G_j(s) = \sum_k O_{k,j}(s)$	network count of bin j up to round s (similarly the reward sum $G_j^S(s)$)
$e(t)$	reference epoch at round t, defined in (2)

Phase 1 (rounds $1, \dots, T_1$). Every agent pulls uniformly at random in rounds $1, \dots, T_1$. After round T_0, agent k floods the record

$$b_k = \sum_{t \le T_0} \varphi_\tau\big(y_{k,t} S(x_{k,t,a_{k,t}})\big) \in \mathbb{R}^d,$$

with τ set for the pooled size $n = NT_0$. By Assumption 2, every agent holds all N records before round $T_1 + 1$. Every agent then computes the same direction

$$\hat{\theta}_0 = \frac{\sum_k b_k}{\|\sum_k b_k\|_1}.$$

Pulls in rounds $T_0 < t \le T_1$ are uniform and are not used.

Bins. An arm x belongs to bin $b(x)$, the bin with $\langle x, \hat{\theta}_0 \rangle \in I_{b(x)}$.

Phase 2: synchronised epochs. Every agent keeps a log of its own Phase-2 pulls, so it can compute $O_{k,\cdot}(s)$ and $O^S_{k,\cdot}(s)$ for any past round s.

(a) *Trigger.* In the communication step of a slot boundary β_m ($m \ge 1$), agent l raises an *alarm* if, for some bin j,

$$\delta_{l,j}(\beta_m) := O_{l,j}(\beta_m) - O_{l,j}(\epsilon_m) \;>\; \gamma \max\big(G_j(\epsilon_m), 1\big), \tag{1}$$

where $\epsilon_m = \max\{e \in \mathcal{E} : e \le \beta_{m-1}\}$. In words: l's own pulls into bin j since the last known epoch exceed γ times the network count at that epoch, which is the count every agent acts on in the next slot. The alarm (one scalar: m) is flooded. If at least one agent raises an alarm at β_m, then $\beta_m \in \mathcal{E}$.

(b) *Snapshot.* When agent k first learns that $\beta_m \in \mathcal{E}$, it floods its *snapshot*: the bins j with $O_{k,j}(\beta_m) \ne O_{k,j}(\epsilon_m)$, each with the pair of increments, plus the epoch label.

(c) *Acting.* At round t in slot $m+1$ (that is, $\beta_m < t \le \beta_{m+1}$), agent i uses the reference epoch

$$e(t) = \max\{e \in \mathcal{E} :\; e \le \beta_{m-1}\}, \tag{2}$$

which is the same for all agents. It acts on the table

$$n_{i,j}(t) = G_j(e(t)) + \big[O_{i,j}(t-1) - O_{i,j}(e(t))\big], \qquad S_{i,j}(t) = G^S_j(e(t)) + \big[O^S_{i,j}(t-1) - O^S_{i,j}(e(t))\big].$$

Among the bins its arms fall into, it picks $J = \arg\max_j \mathrm{UCB}_{i,j}(t)$, where $\mathrm{UCB}_{i,j} = S_{i,j}/n_{i,j} + w(n_{i,j})$ and $w(0) = \infty$, and plays any arm in bin J.

The confidence width is

$$w(n) = c_\delta \sqrt{2/n}, \qquad c_\delta = \sigma \sqrt{2 \log\big(4NN_{\mathrm{b}}(NT)^2/\delta\big)}.$$

Lemma 9 (The protocol is well defined)**.** *(i) In the communication step of β_m, every agent knows ϵ_m and $G(\epsilon_m)$.*

(ii) By the end of round $\beta_m - 1$, every agent knows $G(e), G^S(e)$ for every epoch $e \le \beta_{m-1}$.

Proof. An alarm for $\beta_{m'}$ is flooded in step $\beta_{m'}$ and reaches everyone by step $\beta_{m'} + \Lambda - 1$. An agent that learns of it in step r floods its snapshot in step $r + 1 \le \beta_{m'} + \Lambda$, so every snapshot reaches everyone by step $\beta_{m'} + 2\Lambda - 1 = \beta_{m'+1} - 1$, using $2\Lambda = L$.

For (ii), take $m' \le m - 1$: everything about epoch $\beta_{m'}$ is known by step $\beta_{m'+1} - 1 \le \beta_m - 1$. Every agent can then sum snapshots to get $G(e)$, because $G_j(e) - G_j(e_{\mathrm{prev}})$ is the sum over k of the increments in the snapshots for e.

For (i), whether $\beta_{m'} \in \mathcal{E}$ is known by step $\beta_{m'} + \Lambda - 1 < \beta_m$ for all $m' \le m - 1$, so ϵ_m is known. Since $\epsilon_m \le \beta_{m-1}$, (ii) shows that $G(\epsilon_m)$ is known before round β_m starts. □

Remark 5 (Relation to the code)**.** The flooding code (`src/dec/phase2/flood.py`) works differently. Each agent republishes its own table when its *own* trigger fires, and each agent acts on the freshest copy of every record it holds. That uses more information, but it changes the confidence argument. In the code, whether a pull of agent k is in agent i's view depends on when k next publishes, which is decided *after* the pull. Agent i's view is then a union of N per-agent prefixes. Bounding each prefix separately costs a factor $\sqrt{N}$ in the width (this was the gap in the earlier sketch). The epoch protocol above avoids this: every agent's view is one common prefix plus its own pulls (Lemma 13). E23 measures the code version. E31 implements the epoch version (`src/dec/phase2/epoch.py`): it pays 3–6.5× more extra regret than the code version, because every agent waits for the farthest agent's data. Section 5 analyses a one-line variant of the code version, the *published-only relay*. It has a single-prefix view, so it needs no epochs, and it performs like the code version (E36).

4 Regret upper bound

4.1 Good events and bin geometry

Throughout this section, agent k may use its own direction $\hat{\theta}_k$; the protocol above has $\hat{\theta}_k = \hat{\theta}_0$ for every k. This generality is used again in Section 8. Define the *projection error*

$$\varepsilon_{\mathrm{p}} = X_{\max} \max_{k \le N} \|\hat{\theta}_k - \theta^*\|_1.$$

Lemma 10 (Phase 1 is exactly centralised)**.** *$\sum_k b_k/(NT_0)$ is the truncated Stein estimator on $n = NT_0$ i.i.d. samples. Hence, if NT_0 satisfies the sample-size condition of Lemma 2, there is an event E_θ with $\mathbb{P}(E_\theta) \ge 1 - \delta$ on which $\|\hat{\theta}_0 - \theta^*\|_1 \le \varepsilon_0(NT_0)$.*

Proof. Truncation φ_τ acts on each summand separately, with the same τ at every agent, so $\sum_k b_k = \sum_k \sum_{t \le T_0} \varphi_\tau(y_{k,t} S(x_{k,t}))$ is the sum over the pooled sample. Each pull is uniform over the K i.i.d. contexts, and the noise is i.i.d. (Assumption 1(i)), so $(x_{k,t}, y_{k,t})$ are i.i.d. across (k,t) with the law of a single-agent Phase-1 sample. Apply Lemma 2 with $n = NT_0$. □

Let $\tilde{I}_j = \big(I_j + [-\varepsilon_{\mathrm{p}}, \varepsilon_{\mathrm{p}}]\big) \cap [-X_{\max}, X_{\max}]$ be bin j widened by ε_{p} on each side, and

$$m_j = \sup_{z \in \tilde{I}_j} f(z), \qquad \omega = L_{f'}(\Delta + 2\varepsilon_{\mathrm{p}}).$$

Lemma 11 (Bin geometry)**.** *On E_x, for every agent k and every arm x it sees:*
(i) x falls in some bin, i.e. $|\langle x, \hat{\theta}_k\rangle| \le W$;
(ii) if x is in bin j under $\hat{\theta}_k$, then its true index $z = \langle x, \theta^*\rangle$ lies in $\tilde{I}_j$;
(iii) hence $m_j - \omega \le f(z) \le m_j$.

Proof. (i) $|\langle x, \hat{\theta}_k\rangle| \le \|x\|_\infty \|\hat{\theta}_k\|_1 = \|x\|_\infty \le X_{\max} = W$. (ii) By Hölder, $|\langle x, \hat{\theta}_k\rangle - z| \le \|x\|_\infty \|\hat{\theta}_k - \theta^*\|_1 \le \varepsilon_{\mathrm{p}}$. Also $|z| \le \|x\|_\infty \|\theta^*\|_1 \le X_{\max}$. (iii) $f(z) \le m_j$ by definition. The interval $\tilde{I}_j$ has length at most $\Delta + 2\varepsilon_{\mathrm{p}}$, and f is $L_{f'}$-Lipschitz on it, so $f(z) \ge m_j - \omega$. □

4.2 Confidence

For each bin j, list all Phase-2 pulls into bin j in lexicographic order, and let $\mu_s = f(z_s)$ and η_s be the mean and the noise of the s-th such pull.

Lemma 12 (What an agent sees)**.** *The pulls counted in $n_{i,j}(t)$ are the union of two sets:*

(i) a common prefix*: all pulls into bin j in rounds $\le e(t)$, i.e. the first $G_j(e(t))$ pulls of bin j's lexicographic list;*

(ii) an own segment*: agent i's own pulls into bin j in rounds $e(t) < t' \le t-1$, i.e. a contiguous block of agent i's own bin-j pulls.*

Proof. Immediate from the definition of $n_{i,j}(t)$ and $G_j(e) = \sum_k O_{k,j}(e)$. A pull in round $t' \le e(t)$ precedes, in lexicographic order, every pull in a later round. □

Lemma 13 (Confidence)**.** *There is an event E_η with $\mathbb{P}(E_\eta) \ge 1-\delta$ on which, for every agent i, bin j and Phase-2 round t with $n_{i,j}(t) \ge 1$,*

$$\Big|\frac{S_{i,j}(t)}{n_{i,j}(t)} - \frac{1}{n_{i,j}(t)} \sum_{s \in \mathrm{seen}_{i,j}(t)} \mu_s\Big| \le w\big(n_{i,j}(t)\big),$$

where $\mathrm{seen}_{i,j}(t)$ is the set of pulls counted in $n_{i,j}(t)$.

Proof. By Lemma 12, $S_{i,j} - \sum_{\mathrm{seen}} \mu_s = P + Q$, where P is the noise sum over a prefix of length p of bin j's list, and Q is the noise sum over a contiguous block of length q of agent i's own bin-j pulls, with $p + q = n_{i,j}$.

Prefixes. Let ξ_m be the noise of the m-th pull into bin j. Let $\mathcal{G}_m$ be the information up to and including the *choice* of that pull, and $\mathcal{H}_m = \mathcal{G}_m \vee \sigma(\xi_m)$ the information up to and including its *reward*. Then ξ_m is $\mathcal{H}_m$-measurable and $\mathcal{H}_{m-1} \subseteq \mathcal{G}_m$. The m-th pull happens at a random position (t,i), but on the event $\{$the m-th pull is at $(t,i)\}$, which is decided by $(\mathcal{F}_{t,i}, a_{i,t})$, we have $\xi_m = \eta_{i,t}$, and $\eta_{i,t}$ is independent of $(\mathcal{F}_{t,i}, a_{i,t})$ by Assumption 1(i). Splitting on the position gives $\mathbb{E}[e^{\lambda\xi_m} \mid \mathcal{G}_m] = \mathbb{E}[e^{\lambda\eta}] \le e^{\lambda^2\sigma^2/2}$, and the tower property gives the same bound given $\mathcal{H}_{m-1}$. (This is the stack-of-rewards argument of Lattimore & Szepesvári, *Bandit Algorithms*, Sec. 4.6.) If bin j receives fewer than m pulls, pad the sequence with independent $\mathcal{N}(0,\sigma^2)$ variables; this changes no actual sum. Lemma 4 applies for each fixed $p \le NT$.

Own blocks. The same argument applies to the subsequence of agent i's bin-j pulls, and to the sum from its $(a+1)$-th to its b-th element, for each fixed pair $0 \le a < b \le T$.

Union bound. There are at most $N_{\mathrm{b}} \cdot NT$ prefixes and $N \cdot N_{\mathrm{b}} \cdot T^2$ own blocks, in total at most $2NN_{\mathrm{b}}(NT)^2$ events. Applying Lemma 4 to each with $\delta' = \delta/(2NN_{\mathrm{b}}(NT)^2)$, with probability $\ge 1-\delta$ all of them satisfy

$$|P| \le c_\delta\sqrt{p}, \qquad |Q| \le c_\delta\sqrt{q}.$$

Since $\sqrt{p}+\sqrt{q} \le \sqrt{2(p+q)}$, we get $|P+Q| \le c_\delta\sqrt{2n_{i,j}}$. Dividing by $n_{i,j}$ gives the claim. □

4.3 Regret of one pull

Let $E = E_x \cap E_\theta \cap E_\eta$. Then $\mathbb{P}(E^c) \le 3\delta$.

Lemma 14 (Per-pull regret)**.** *On $E_x \cap E_\eta$, if agent i pulls from bin J at Phase-2 round t, then*

$$r_{i,t} \le \min\big\{2L_f,\ 2w(n_{i,J}(t)) + 2\omega\big\}.$$

This holds for any directions $\hat\theta_k$ (all equal or not) and for any set of available bins.

Proof. $r_{i,t} \le 2L_f$ because $|f| \le L_f$ on E_x. For the second bound, assume $n_{i,J} \ge 1$ (otherwise $w = \infty$). Let x^* be the best arm, in bin j^* under $\hat\theta_i$. By Lemma 11, $f(\langle x^*, \theta^*\rangle) \le m_{j^*}$.

The best bin looks good. If $n_{i,j^*} = 0$, then $\mathrm{UCB}_{j^*} = \infty$. Since J maximises the UCB, $\mathrm{UCB}_J = \infty$, contradicting $n_{i,J} \geq 1$. So $n_{i,j^*} \geq 1$. Every seen pull of bin j^* has mean $\geq m_{j^*} - \omega$ (Lemma 11), so by Lemma 13,

$$\mathrm{UCB}_{j^*} \;\geq\; (m_{j^*} - \omega) - w(n_{i,j^*}) + w(n_{i,j^*}) \;=\; m_{j^*} - \omega.$$

The chosen bin is not much better than it looks. Every seen pull of bin J has mean $\leq m_J$, so

$$\mathrm{UCB}_J \;\leq\; m_J + 2w(n_{i,J}).$$

Combine. The pulled arm has mean $\geq m_J - \omega$, and $\mathrm{UCB}_J \geq \mathrm{UCB}_{j^*}$. Therefore

$$r_{i,t} \;\leq\; m_{j^*} - (m_J - \omega) \;\leq\; (\mathrm{UCB}_{j^*} + \omega) - (\mathrm{UCB}_J - 2w(n_{i,J})) + \omega \;\leq\; 2w(n_{i,J}) + 2\omega. \qquad \square$$

4.4 Stale counts

This is the only place where the graph enters the regret bound.

Lemma 15 (Unseen pulls)**.** *Let t be a Phase-2 round, J a bin and $l \neq i$ an agent. The number of pulls by l into J in rounds $e(t) < t' \leq t-1$ is at most $\gamma \max(G_J(e(t)), 1) + 2L - 1$.*

Proof. Let t be in slot $m+1$, so $\beta_m < t \leq \beta_{m+1}$ and $e(t) = \max\{e \in \mathcal{E} : e \leq \beta_{m-1}\}$. Split the rounds $(e(t), t-1]$ into $(e(t), \beta_{m-1}]$ and $(\beta_{m-1}, t-1]$.

Second part. It has $t - 1 - \beta_{m-1} \leq \beta_{m+1} - 1 - \beta_{m-1} = 2L - 1$ rounds, and l pulls once per round.

First part. If $e(t) = \beta_{m-1}$, it is empty; this covers $m \leq 1$. Otherwise $\beta_{m-1} \notin \mathcal{E}$, so no agent raised an alarm at β_{m-1}. Here $\epsilon_{m-1} = \max\{e \in \mathcal{E} : e \leq \beta_{m-2}\}$, and because $\beta_{m-1} \notin \mathcal{E}$ this equals $\max\{e \in \mathcal{E} : e \leq \beta_{m-1}\} = e(t)$. So (1) failing for l and J at β_{m-1} reads

$$O_{l,J}(\beta_{m-1}) - O_{l,J}(e(t)) \;\leq\; \gamma \max\big(G_J(e(t)), 1\big),$$

which is the claim for the rounds $(e(t), \beta_{m-1}]$. $\square$

Lemma 16 (Stale counts)**.** *If the pull of agent i at round t is the k-th Phase-2 pull into bin J (in lexicographic order), then*

$$n_{i,J}(t) \;\geq\; \frac{k - c_{N,\Lambda}}{1 + N\gamma}, \qquad c_{N,\Lambda} := 1 + N(\gamma + 2L) = 1 + N(\gamma + 4\Lambda).$$

Proof. Write $n = n_{i,J}(t)$ and $G = G_J(e(t)) \leq n$. The $k-1$ earlier pulls into J are of three kinds:

- seen by agent i: exactly n of them (Lemma 12);
- by another agent $l \neq i$ in rounds $(e(t), t-1]$: at most $(N-1)\big(\gamma \max(G,1) + 2L - 1\big)$ by Lemma 15;
- by another agent in round t itself: at most $N - 1$.

So $k - 1 \leq n + (N-1)\gamma \max(G, 1) + (N-1)\,2L$.

Case $G \geq 1$. Then $\max(G,1) = G \leq n$, so $k - 1 \leq n(1 + N\gamma) + 2NL$, and $n \geq (k - 1 - 2NL)/(1 + N\gamma) \geq (k - c_{N,\Lambda})/(1 + N\gamma)$.

Case $G = 0$. Then $k - 1 \leq n + N(\gamma + 2L)$, so $n \geq k - c_{N,\Lambda}$. If $k \leq c_{N,\Lambda}$ the claim is trivial. Otherwise $n \geq k - c_{N,\Lambda} \geq (k - c_{N,\Lambda})/(1 + N\gamma)$. $\square$

4.5 The regret theorem

Theorem 17 (Regret of Dec-ZoomSIB-Relay)**.** *Let Assumptions 1 and 2 hold, and let $\delta \in (0,1)$. Suppose NT_0 satisfies the sample-size condition of Lemma 2. Then for any $\Delta > 0$ and $\gamma > 0$, with probability at least $1 - 3\delta$,*

$$R_T(N) \;\le\; \underbrace{2L_f N(T_0+\Lambda)}_{\text{Phase 1}} + \underbrace{4\sqrt{2}\,c_\delta\sqrt{(1+N\gamma)N_{\mathrm{b}}NT}}_{\text{UCB}} + \underbrace{2L_{f'}\Delta NT}_{\text{discretisation}} + \underbrace{4L_{f'}X_{\max}\,\varepsilon_0(NT_0)\,NT}_{\text{projection}} + \underbrace{2L_f N_{\mathrm{b}}\,(c_{N,\Lambda}+1)}_{\text{delay and staleness}}.$$

Proof. Idea. Each Phase-2 pull costs at most twice the confidence width of its bin, plus the bin's width (Lemma 14). The graph only makes the count an agent sees lag the true count, by a factor $1 + N\gamma$ and an additive $c_{N,\Lambda}$ (Lemma 16). Summing the widths over the pulls gives the server bound plus the cost of the lag.

We work on E, which has $\mathbb{P}(E) \ge 1 - 3\delta$.

Phase 1. There are $NT_1 = N(T_0+\Lambda)$ uniform pulls, each costing at most $2L_f$ on E_x.

Phase 2 on E. On E_θ, $\varepsilon_{\mathrm{p}} \le X_{\max}\varepsilon_0(NT_0)$. Fix a bin j and let n_j be its final number of Phase-2 pulls. For the k-th of them, Lemmas 14 and 16 give

$$r^{(k)} \;\le\; \min\Big\{2L_f,\; 2c_\delta\sqrt{\tfrac{2(1+N\gamma)}{k-c_{N,\Lambda}}} + 2\omega\Big\},$$

where the second term is used only when $k > c_{N,\Lambda}$. Use $2L_f$ for $k \le c_{N,\Lambda}+1$; at most $c_{N,\Lambda}+1$ such terms. For $k > c_{N,\Lambda}+1$, the numbers $k - c_{N,\Lambda}$ are at least 1 and spaced by 1, so $\sum_k (k-c_{N,\Lambda})^{-1/2} \le \sum_{u=1}^{n_j} u^{-1/2} \le 2\sqrt{n_j}$. Hence

$$\sum_{k=1}^{n_j} r^{(k)} \;\le\; 2L_f(c_{N,\Lambda}+1) + 4\sqrt{2}\,c_\delta\sqrt{(1+N\gamma)\,n_j} + 2\omega\,n_j.$$

Sum over the N_{b} bins. Cauchy–Schwarz gives $\sum_j \sqrt{n_j} \le \sqrt{N_{\mathrm{b}}\sum_j n_j} \le \sqrt{N_{\mathrm{b}}NT}$. Finally, $2\omega NT = 2L_{f'}\Delta NT + 4L_{f'}\varepsilon_{\mathrm{p}}NT$. □

Remark 6 (In expectation)**.** Assumption 1 bounds f only on $|z| \le X_{\max}$, so off E_x the regret is not controlled. For this reason the theorem is stated with high probability, as DBG state their upper bound. If $|f| \le L_f$ on all of $\mathbb{R}$, then $R_T(N) \le 2L_f NT$ always, and $\delta = 1/(NT)$ gives $\mathbb{E}R_T(N) \le$ (the bound above)$+6L_f$. This holds on the hard instances of Lemma 3 (where $|f| \le 1$), and it is the form we compare with Theorem 5.

Corollary 18 (Rate)**.** *Take $\delta = 1/(NT)$, $\Delta = (NT)^{-1/3}$, $\gamma = \gamma_0/N$ for a constant γ_0, and*

$$NT_0 = \left\lceil \left(L_{f'}X_{\max}\tfrac{4C_\theta d}{\mu^*}\sqrt{\log(2dNT)}\,\tfrac{NT}{L_f}\right)^{2/3}\right\rceil$$

(the choice of the server theorem), at least as large as Lemma 2 requires. Then, with probability at least $1 - 3/(NT)$,

$$R_T(N) = \tilde{O}\Big((d\,NT)^{2/3} + N\Lambda\,(NT)^{1/3}\Big).$$

If $|f| \le L_f$ on $\mathbb{R}$, the same holds for $\mathbb{E}R_T(N)$ (Remark 6). If also $N\Lambda \le (NT)^{1/3}$, this matches Theorem 5 in N and T up to logarithms.

Proof. $N_{\mathrm{b}} \le 2X_{\max}(NT)^{1/3} + 1 = \tilde{O}((NT)^{1/3})$, and $c_\delta = \tilde{O}(1)$, so with $N\gamma = \gamma_0$ the UCB term is $\tilde{O}((NT)^{2/3})$. The discretisation term is $2L_{f'}(NT)^{2/3}$. The choice of T_0 equalises $2L_f NT_0$ and the projection term $\propto d\,NT/\sqrt{NT_0}$, and makes both $\tilde{O}((dNT)^{2/3})$. The Phase-1 delay $2L_f N\Lambda$ is dominated by the last term. The last term is $2L_f N_{\mathrm{b}}(2 + \gamma_0 + 4N\Lambda) = \tilde{O}(N\Lambda(NT)^{1/3})$. □

Remark 7 (What changed from the sketch)**.** The earlier sketch (`server.tex`, Theorem 16) bounded the confidence width of an agent's view, which is a union of N per-agent prefixes, and lost a factor $\sqrt{N}$. Here the view is a common prefix plus one own block (Lemma 12), so Lemma 13 is exactly the server lemma. The price is that agents act on slightly older data: the delay constant is 4Λ instead of Λ. This is the "delayed but consistent statistics" idea used by Dubey & Pentland (2020) and by DLUCB.

4.6 A sharper delay term

The proof of Theorem 17 charges the full $2L_f$ to each of the first $c_{N,\Lambda} + 1$ pulls of a bin. That is wasteful: an agent always sees its *own* earlier pulls, and the first $c_{N,\Lambda} + 1$ pulls are shared among N agents. Each agent therefore pays a UCB-type cost on its own share, and only its very first pull is blind.

Theorem 19 (Sharper delay term)**.** *Under the conditions of Theorem 17, with probability at least $1 - 3\delta$ the bound of Theorem 17 holds with the delay term $2L_f N_{\mathrm{b}}(c_{N,\Lambda} + 1)$ replaced by*

$$N_{\mathrm{b}} \cdot \min\Big\{2L_f(c_{N,\Lambda} + 1),\quad 2L_f N + 4\sqrt{2}\,c_\delta\sqrt{N(c_{N,\Lambda} + 1)}\Big\}.$$

Proof. Work on E and fix a bin j. Write $c = c_{N,\Lambda}$. The proof of Theorem 17 bounds the pulls with index $k > c + 1$ exactly as before, and charges $2L_f$ to each of the first $c + 1$ pulls. We bound the first $c + 1$ pulls in a second way and keep the smaller bound.

Own pulls are always seen. Let agent i make a pull into j at round t. Since $G_j(e) \ge O_{i,j}(e)$,

$$n_{i,j}(t) = G_j(e(t)) + O_{i,j}(t-1) - O_{i,j}(e(t)) \;\ge\; O_{i,j}(t-1),$$

the number of i's earlier pulls into j.

Split the first $c+1$ pulls by agent. Let m_i be the number of them made by agent i, so $\sum_i m_i \le c + 1$. They are agent i's first m_i pulls into j, so the s-th of them ($s = 0, \dots, m_i - 1$) has $n_{i,j} \ge s$. By Lemma 14 it costs at most $2L_f$ if $s = 0$, and at most $2c_\delta\sqrt{2/s} + 2\omega$ if $s \ge 1$. Using $\sum_{s=1}^{m} s^{-1/2} \le 2\sqrt{m}$ and Cauchy–Schwarz over the at most N agents with $m_i \ge 1$,

$$\begin{aligned}\sum_{\text{first } c+1 \text{ pulls}} r \;&\le\; 2L_f \min(N, c+1) + 4\sqrt{2}\,c_\delta \sum_i \sqrt{m_i} + 2\omega(c+1)\\ &\le\; 2L_f N + 4\sqrt{2}\,c_\delta\sqrt{N(c+1)} + 2\omega(c+1).\end{aligned}$$

The 2ω terms join the discretisation and projection terms as before. Sum over the N_{b} bins. □

Corollary 20 (Rate with the sharper delay term)**.** *With the choices of Corollary 18, with probability at least $1 - 3/(NT)$,*

$$R_T(N) = \tilde{O}\Big((d\,NT)^{2/3} + N\sqrt{\Lambda}\,(NT)^{1/3}\Big).$$

This matches Theorem 5 in N and T up to logarithms whenever $N\sqrt{\Lambda} \le (NT)^{1/3}$, a wider range than the $N\Lambda \le (NT)^{1/3}$ of Corollary 18.

Proof. With $\gamma = \gamma_0/N$, $c_{N,\Lambda} + 1 = 2 + \gamma_0 + 4N\Lambda \le N(2 + \gamma_0 + 4\Lambda)$, so $\sqrt{N(c_{N,\Lambda}+1)} \le N\sqrt{2+\gamma_0+4\Lambda}$. The delay term is then at most

$$N_{\mathrm{b}} N\big(2L_f + 4\sqrt{2}\, c_\delta \sqrt{2+\gamma_0+4\Lambda}\big) = \tilde{O}\big(N\sqrt{\Lambda}\,(NT)^{1/3}\big),$$

using $N_{\mathrm{b}} = \tilde{O}((NT)^{1/3})$ and $c_\delta = \tilde{O}(1)$. The other terms are as in Corollary 18. □

4.7 A delay term that does not grow with the horizon

Theorems 17 and 19 let *every* bin have up to $c_{N,\Lambda}$ unseen pulls, so their delay term carries the factor $N_{\mathrm{b}} \sim (NT)^{1/3}$. But a pull into bin j needs an arm in bin j, and each agent sees only K arms per round. Summed over all bins, the number of recent unseen pulls is about $N\Lambda K$, whatever N_{b} and T are.

Let $q_j = \mathbb{P}$(at least one of the K arms of a round falls in bin j), computed with the frozen direction. Since $\sum_j q_j$ is the expected number of distinct bins among K arms,

$$\textstyle\sum_j q_j \;\le\; \min(K, N_{\mathrm{b}}).$$

Lemma 21 (Availability in short windows)**.** *Let $W \ge 1$ and $\ell = \log(TN_{\mathrm{b}}/\delta)$. With probability at least $1-\delta$, for every bin j and every window of W consecutive Phase-2 rounds, bin j is available at no more than $2NWq_j + 3\ell$ of the NW agent-rounds of the window.*

Proof. Given the frozen direction, which depends only on Phase-1 data, the Phase-2 contexts are fresh. So the indicators "bin j is available to agent k at round r" are independent Bernoulli(q_j). For a sum S of independent Bernoulli variables with mean μ, the multiplicative Chernoff bound $\mathbb{P}(S \ge (1+a)\mu) \le e^{-a^2\mu/(2+a)}$ with $a = 1 + 3\ell/\mu$ gives $\mathbb{P}(S \ge 2\mu + 3\ell) \le e^{-\ell}$. A union bound over at most T windows and N_{b} bins finishes the proof. □

Theorem 22 (Horizon-free delay term)**.** *Under the conditions of Theorem 17, with probability at least $1-4\delta$, the bound of Theorem 17 holds with the UCB term replaced by $8c_\delta\sqrt{(1+N\gamma)N_{\mathrm{b}}NT}$ and the delay term replaced by*

$$\big(2L_f + 4\sqrt{2}\, c_\delta\big)\Big(16\, N\Lambda \min(K, N_{\mathrm{b}}) \;+\; N_{\mathrm{b}}\,(6\ell + N\gamma + 1)\Big).$$

The first part is the only one that depends on the graph, and it does not depend on T or N_{b}. The second part does not depend on N or Λ.

Proof. Idea. A pull is *late* if its bin has already received many more pulls than can be unseen at that moment. Late pulls see at least a constant fraction of the true count, which costs only a constant in the UCB term. The *early* pulls of a bin are few, and their number is set by how often the bin is available in a short window.

Work on E and on the event of Lemma 21 with $W = 2L$. Let agent i make the k-th pull into bin J at round t in slot $m+1$, and let $n = n_{i,J}(t)$ and $G = G_J(e(t)) \le n$. Recall the proof of Lemma 16. The earlier pulls that agent i does not see are, for each other agent, at most $\gamma \max(G, 1)$ pulls in $(e(t), \beta_{m-1}]$ (the failed trigger), plus the other agents' pulls into J in rounds $(\beta_{m-1}, t]$. Call the latter number u. These rounds form a window of at most $2L$ rounds, so $u \le U_J := 4NLq_J + 3\ell$. Hence

$$k - 1 \;\le\; n + N\gamma \max(n, 1) + u.$$

Late pulls, those with $k - 1 > 2U_J + N\gamma$. If $n = 0$, the display gives $k - 1 \le N\gamma + U_J$, which is impossible. So $n \ge 1$, and $(1+N\gamma)n \ge k - 1 - U_J \ge (k-1)/2$. By Lemma 14 the pull costs at most $4c_\delta\sqrt{(1+N\gamma)/(k-1)} + 2\omega$. Summing over $k \ge 2$ gives at most $8c_\delta\sqrt{(1+N\gamma)n_J} + 2\omega n_J$.

Early pulls, the first $V_J := \lfloor 2U_J + N\gamma \rfloor + 1$ pulls of bin J. As in the proof of Theorem 19, agent i's early pulls are its first m_i pulls into J, and the s-th of them sees $n \geq s$. Their total cost is at most $2L_f \#\{i : m_i \geq 1\} + 4\sqrt{2}\, c_\delta \sum_i \sqrt{m_i} + 2\omega V_J \leq (2L_f + 4\sqrt{2}\, c_\delta)V_J + 2\omega V_J$, using $\sqrt{m_i} \leq m_i$ and $\sum_i m_i \leq V_J$.

Sum over bins. $\sum_J V_J \leq \sum_J (8NLq_J + 6\ell + N\gamma + 1) \leq 8NL \min(K, N_\mathrm{b}) + N_\mathrm{b}(6\ell + N\gamma + 1)$, with $L = 2\Lambda$. The late pulls give the UCB term after Cauchy–Schwarz, and the 2ω terms give the discretisation and projection terms, as in Theorem 17. □

Corollary 23 (Rate with a horizon-free delay term)**.** *With the choices of Corollary 18, with probability at least* $1 - 4/(NT)$*,*

$$R_T(N) = \tilde{O}\Big((d\,NT)^{2/3} + N\Lambda \min(K, N_\mathrm{b})\Big).$$

For a fixed number of arms, the graph costs an additive term that does not grow with the horizon*. This matches Theorem 5 whenever* $N\Lambda K \leq (NT)^{2/3}$*.*

Proof. The term $N_\mathrm{b}(6\ell + \gamma_0 + 1)$ is $\tilde{O}((NT)^{1/3})$ and independent of the graph. The rest is as in Corollary 18. □

Remark 8 (Comparison with the measurements)**.** Theorems 19 and 22 are both valid; one can use the smaller of the two delay terms. For a fixed K, Theorem 22 gives the shape that E23 measures: flat in T. For the epoch protocol it predicts $\propto N\Lambda K$, and in E31/E36 the measured excess divided by NDK is between 0.86 and 1.15 on all five graphs. In E36 an 8× longer horizon raises the excess only by ×1.2 (the log factors). The excess grows like $K^{0.44}$, not linearly: the factor K comes from bounding availability by $\sum_j q_j \leq K$, and it is loose.

Remark 9 (The trigger threshold)**.** The factor $\sqrt{1 + N\gamma}$ appears for the same reason as with a server: between two epochs, each of the N agents may add $\gamma\, G_J$ pulls to bin J. E30 (server) and E31 (flooding, both protocols) find $\gamma \leq 0.5$ free in practice: within 2% of $\gamma = 1/N$ on the torus and the ring. Only $\gamma = 2$ costs, ×2–3.

5 A relay we can analyse: the published-only relay

The epoch protocol makes every agent wait for the farthest agent, and E31 measures the price: 3–6.5× the extra regret of the flood relay in the code. The flood relay's proof broke because a pull's reward could change *when* that pull is published. Its agent uses its own unpublished rewards, which change its next pulls, which change when its trigger fires. One change removes the problem: *every agent, the publisher included, acts only on published records.*

Protocol (Phase 2). Phase 1 is as in Section 3. Each agent k keeps its own counts and sums. In the communication step of round s it *publishes*, i.e. floods a versioned copy of its own table, if for some bin j

$$(\text{own pulls into } j \text{ not yet published}) \;>\; \gamma \max\big(V_{k,j}(s), 1\big), \tag{3}$$

where $V_{k,j}(s)$ is the count agent k acts on at round s. In this section the graph is static and connected, and records are flooded one hop per step as in Section 1. So a record published by l in step s can be used by agent i from round $s + \max(d(i,l), 1)$ on, and not earlier (Lemma 1; the $\max(\cdot, 1)$ is the publisher's own one-round delay). Agent i acts on the sum, over all agents l, of the freshest record of l it can use, so $n_{i,j}(t) = V_{i,j}(t)$. This is the code's flood relay (`src/dec/phase2/flood.py`) with `own=False`. Any $\gamma > 0$ is allowed.

Lemma 24 (Rewards travel only inside published records)**.** *Let pull m be published in step s_m. If its reward influences anything agent k' does at round or step τ (an action or a trigger decision), then $\tau \geq s_m + \max(d(k_m, k'), 1)$.*

Proof. Agents decide actions from their usable records and their contexts, and triggers from counts. Counts depend on actions only, never directly on rewards. So a reward can enter a decision only through a record that carries it, or carries something it changed. We show by induction on τ: if y_m changes something agent k' does at time τ, then $\tau \geq s_m + \max(d(k_m, k'), 1)$. The first thing y_m can change is a decision of an agent that has received a record containing y_m itself; such a record left k_m in step s_m or later and arrives no earlier than stated. This includes $k' = k_m$, which uses its own pull only once it is published. Otherwise the change reaches k' through a record of some agent a whose content y_m changed at a time $\tau_a \geq s_m + \max(d(k_m, a), 1)$ (induction), and that record arrives at k' no earlier than $\tau_a + \max(d(a, k'), 1)$. By the triangle inequality and $\max(x, 1) + \max(y, 1) \geq \max(x + y, 1)$, this is at least $s_m + \max(d(k_m, k'), 1)$. □

For agent i, give each pull m the key $\kappa_i(m) = \big(s_m + \max(d(i, k_m), 1),\ s_m,\ k_m,\ r_m\big)$, ordered lexicographically, with $s_m = \infty$ if m is never published. Agent i's view at round t is exactly the set of pulls whose first key coordinate is at most t, so it is a *prefix* of the pulls ordered by κ_i.

Lemma 25 (One prefix, no $\sqrt{N}$)**.** *For every agent i and bin j, the noises of the bin-j pulls, taken in the order of κ_i, form a martingale difference sequence with σ-sub-Gaussian increments. Hence there is an event E'_η with $\mathbb{P}(E'_\eta) \geq 1 - \delta$ on which, for all i, j, t with $n_{i,j}(t) \geq 1$,*

$$\Big|\frac{S_{i,j}(t)}{n_{i,j}(t)} - \frac{1}{n_{i,j}(t)} \sum_{s \in \mathrm{seen}_{i,j}(t)} \mu_s\Big| \leq \frac{c'_\delta}{\sqrt{n_{i,j}(t)}}, \qquad c'_\delta = \sigma\sqrt{2\log\big(2NN_{\mathrm{b}}NT/\delta\big)}.$$

Proof. Influence moves forward in the order. Suppose the reward of pull m influences a decision about pull m' (its action, or the trigger that publishes it), taken at time $\tau \leq s_{m'}$. By Lemma 24, the triangle inequality $d(i, k_m) \leq d(i, k') + d(k', k_m)$, and $\max(a, 1) + \max(b, 1) \geq \max(a + b, 1)$,

$$s_{m'} + \max(d(i, k'), 1) \ \geq\ s_m + \max(d(k_m, k'), 1) + \max(d(i, k'), 1) \ \geq\ s_m + \max(d(i, k_m), 1).$$

If the first coordinates are equal, then $s_{m'} \geq \tau \geq s_m + 1 > s_m$. So in both cases $\kappa_i(m') > \kappa_i(m)$.

Martingale. Order all pulls, of every bin, by κ_i. By the step above, which pull comes next, and whether it is in bin j, is decided by rewards that come earlier in this order (and by contexts and internal randomness). This includes the absence of pulls: whether some pull gets a key below a given value x is decided by actions and trigger decisions taken at times whose κ_i-key is at most x (ties broken by s, as above), and these depend only on rewards with smaller keys. So no pull can later turn out to belong before one already in the sequence. The next pull's own reward is not among them. With the noises drawn in advance (the stack-of-rewards model of Lattimore & Szepesvári, Sec. 4.6), the next noise is therefore independent of everything that came before in the order, and has the law of η. Restricting to bin j is optional skipping, which keeps the martingale property.

Union bound. The view is a prefix of this sequence (one prefix, no own block). Lemma 4 for every agent, bin and prefix length $p \leq NT$, with $\delta' = \delta/(NN_{\mathrm{b}}NT)$, gives the claim. □

Lemma 26 (Stale counts for the published-only relay)**.** *Let $F' = \max_i \sum_l \max(d(i, l), 1)$, which is the largest farness $\max_i \sum_l d(i, l)$ plus 1 (only $l = i$ has $d = 0$), let $\ell' = \log(NN_{\mathrm{b}}T/\delta)$ and*

$U'_J = 2F'q_J + 3\ell'$. *With probability at least* $1-\delta$ *the following holds for every pull: if agent* i *makes the* k*-th pull into bin* J *(in lexicographic order) at round* t*, and* $n = n_{i,J}(t)$*, then*

$$k-1 \;\le\; n + N\gamma \max(n,1) + U'_J.$$

In particular, if $k-1 > 2U'_J + N\gamma$*, then* $n \ge 1$ *and* $n \ge \dfrac{k-1}{2(1+N\gamma)}$.

Proof. Step 1: views nest. For all agents i, l and every round s, every record that agent l can use at round s, agent i can use at round $s + \max(d(i,l),1)$. Indeed, let a record of agent k be published in step s_k with $s_k + \max(d(l,k),1) \le s$. By the triangle inequality $d(i,k) \le d(i,l) + d(l,k)$ and $\max(a+b,1) \le \max(a,1) + \max(b,1)$,

$$s_k + \max(d(i,k),1) \;\le\; s_k + \max(d(l,k),1) + \max(d(i,l),1) \;\le\; s + \max(d(i,l),1).$$

So for each k, agent i acts at that round on a record of k at least as recent as the one l acts on at round s. Records are cumulative tables, hence

$$V_{l,j}(s) \;\le\; n_{i,j}\big(s + \max(d(i,l),1)\big) \qquad \text{for every bin } j.$$

Step 2: count the unseen pulls. Of the $k-1$ earlier pulls into J, agent i sees exactly n. Take another earlier pull, by an agent l ($l = i$ is allowed: i does not see its own unpublished pulls). Let $s_l = t - \max(d(i,l),1)$. At round t agent i uses l's freshest record published in a step $\le s_l$, so an unseen pull of l is of one of two kinds.

(a) Made in a round $\le s_l$*, but still unpublished after step* s_l. In step s_l agent l either published, leaving no unpublished pulls, or its trigger (3) failed. Either way, after step s_l it has at most $\gamma \max(V_{l,J}(s_l),1)$ unpublished pulls in J. By Step 1 with $s = s_l$, $V_{l,J}(s_l) \le n_{i,J}(t) = n$. So this kind numbers at most $\gamma\max(n,1)$ for each l, and at most $N\gamma\max(n,1)$ in total. (If $s_l \le T_1$, there are no Phase-2 pulls of this kind.)

(b) Made in a round in $(s_l, t]$. A pull into J needs an arm in bin J, and only Phase-2 rounds have bin pulls. So this kind numbers at most the number u of agent-rounds (l,r) with $\max(s_l, T_1) < r \le t$ at which bin J is available to l. These agent-rounds form a fixed set of size at most $\sum_l \max(d(i,l),1) \le F'$, all in Phase 2. As in Lemma 21, the availability indicators are independent Bernoulli(q_J) given the frozen direction, so the Chernoff bound and a union bound over the N agents i, the N_{b} bins and the T rounds give $u \le 2F'q_J + 3\ell' = U'_J$ for all of them at once, with probability at least $1-\delta$.

Adding up, $k-1 \le n + N\gamma\max(n,1) + U'_J$. For the last claim, $n = 0$ would give $k-1 \le N\gamma + U'_J$, so $n \ge 1$. Then $(1+N\gamma)\,n \ge k - 1 - U'_J > (k-1)/2$. □

This is the stale-count bound of the epoch protocol (Lemma 16): a factor $1+N\gamma$ and an additive lag. It holds for every γ, because Step 1 measures the cap on l's unpublished pulls against a count that i also sees, not against the true count $k-1$. (An earlier version used $V_{l,J} \le k-1$ and so needed $N\gamma \le 1/2$.)

Theorem 27 (Regret of the published-only relay)**.** *Let Assumption 1 hold, let* $\mathcal{G}$ *be static and connected with diameter* D*, so that* $\Lambda = D$ *(Lemma 1), let* $\gamma > 0$*, and let* NT_0 *satisfy the sample-size condition of Lemma 2. With probability at least* $1 - 4\delta$*,*

$$\begin{aligned} R_T(N) \;\le\; & 2L_f N(T_0 + \Lambda) + 4\sqrt{2}\, c'_\delta \sqrt{(1+N\gamma)N_{\mathrm{b}}NT} + 2L_{f'}\Delta NT + 4L_{f'}X_{\max}\varepsilon_0(NT_0)NT \\ & + 2L_f\Big(4F'\min(K, N_{\mathrm{b}}) + N_{\mathrm{b}}(6\ell' + N\gamma + 1)\Big). \end{aligned}$$

With the choices of Corollary 18, including $\gamma = \gamma_0/N$,

$$R_T(N) = \tilde{O}\Big((d\,NT)^{2/3} + K\,F'\Big), \qquad F' \le N\max(D,1).$$

With the same choices but a constant γ (such as $\gamma = 1/2$) and $\gamma N_{\mathrm{b}} \le T$,

$$R_T(N) = \tilde{O}\Big((d\,NT)^{2/3} + \sqrt{1+N\gamma}\,(NT)^{2/3} + K\,F'\Big).$$

Proof. Work on $E_x \cap E_\theta \cap E'_\eta$ and the event of Lemma 26. Lemma 14 holds with the width $c'_\delta/\sqrt{n}$ of Lemma 25; its proof only uses the confidence bound and the bin geometry. Fix a bin J.

Early pulls. Charge $2L_f$ to each of its first $\lfloor 2U'_J + N\gamma \rfloor + 1$ pulls.

Late pulls, with $k - 1 > 2U'_J + N\gamma$. By Lemma 26, $n \ge (k-1)/(2(1+N\gamma))$, so the pull costs at most $2c'_\delta\sqrt{2(1+N\gamma)/(k-1)} + 2\omega$. Using $\sum_{k=2}^{n_J}(k-1)^{-1/2} \le 2\sqrt{n_J}$, these sum to at most $4\sqrt{2}\,c'_\delta\sqrt{(1+N\gamma)\,n_J} + 2\omega n_J$.

Sum over bins. The early pulls give $2L_f\sum_J(2U'_J + N\gamma + 1) \le 2L_f\big(4F'\min(K,N_{\mathrm{b}}) + N_{\mathrm{b}}(6\ell' + N\gamma + 1)\big)$, using $\sum_J q_J \le \min(K, N_{\mathrm{b}})$. The late pulls give the UCB term after Cauchy–Schwarz, and the 2ω terms give the discretisation and projection terms, as in Theorem 17.

Rates. As in Corollary 18, $N_{\mathrm{b}} = \tilde{O}((NT)^{1/3})$ and $c'_\delta = \tilde{O}(1)$. So the UCB term is $\tilde{O}(\sqrt{1+N\gamma}\,(NT)^{2/3})$, which is $\tilde{O}((NT)^{2/3})$ when $N\gamma = \gamma_0$. The term $2L_f N_{\mathrm{b}} N\gamma$ is of lower order whenever $\gamma N_{\mathrm{b}} \le T$, since then $(N_{\mathrm{b}}N\gamma)^2 \le (1+N\gamma)\,N_{\mathrm{b}}\cdot\gamma N_{\mathrm{b}}N \le (1+N\gamma)N_{\mathrm{b}}NT$, so $N_{\mathrm{b}}N\gamma \le \sqrt{(1+N\gamma)N_{\mathrm{b}}NT}$. Finally, $2L_f N\Lambda = 2L_f ND \le 4L_f F'$: for a pair u, v at distance D, every l has $d(u,l) + d(l,v) \ge D$, so $\sum_l d(u,l) + \sum_l d(v,l) \ge ND$ and $F' \ge ND/2$. So the Phase-1 sharing rounds are absorbed into KF'. □

Remark 10 (What this buys).

- **No $\sqrt{N}$ from the view, no epochs.** Each agent's view is a single prefix, so the confidence width is the server's. The only N-dependence in the UCB term is the trigger's $\sqrt{1+N\gamma}$, as with a server.
- **γ is a regret–communication dial.** Theorem 27 holds for every γ. $\gamma = \gamma_0/N$ gives the rate-optimal bound. A constant γ costs $\sqrt{1+N\gamma}$ on the UCB term of the bound, and in return each agent publishes about N times less often, because $\log_{1+\gamma} NT \approx N\log(NT)/\gamma_0$ for $\gamma = \gamma_0/N$.
- **The graph term is horizon-free and smaller by a constant, not by an order.** It is $8L_f F'\min(K, N_{\mathrm{b}})$, against $16(2L_f + 4\sqrt{2}c_\delta)N\Lambda\min(K, N_{\mathrm{b}})$ for the epoch protocol (Theorem 22). On every connected graph $ND/2 \le F' \le N\max(D,1)$ (proof of Theorem 27), so with $\Lambda = D$ both are of order NDK: on a ring $F' \approx ND/2$, on a star $F' \approx 2N = ND$. The gain is a constant (at least 4) and, in practice, the absence of synchronisation: no slots, no alarms, no waiting for the farthest agent. E31 measures 3–6.5× less extra regret than the epoch protocol.
- **Communication.** Between two of agent l's triggers for bin j, its count $V_{l,j}$ grows by a factor $> 1+\gamma$, so each agent publishes at most $N_{\mathrm{b}}(1 + \log_{1+\gamma} NT)$ times. That gives the same bound as Theorem 29, for the same γ. If each publication carries only the increment since the last one, an agent stores $O(N_{\mathrm{b}} + N)$ numbers, as in Proposition 30(i).
- **Measured (E36).** At $\gamma = 0.5$ it matches the flood relay of E23 in regret and in communication: extra regret over the server 538 ± 29 against 528 ± 26 on the ring and 643 ± 22 against 633 ± 23 on the path, with 1.02–1.06× the scalars. It is flat in T (ring, $T = 2{,}500 \to 20{,}000$: $413 \to 545$, against $399 \to 531$). Across K it tracks the flood relay up to $K = 20$, and is worse at $K = 40$ (735 ± 60 against 627 ± 45). At $\gamma = 1/(2N)$ it has *less* extra regret on every graph (complete: -25 ± 6; ring: 381; path: 488), for 7–8× more scalars, so the dial

is visible in practice too, but the regret side of it is far smaller than $\sqrt{1+N\gamma} = 3$ suggests (7,385 against 7,229 total regret on the ring, +2%). Since the theorem now covers $\gamma = 0.5$, the protocol we prove can be the protocol we run, at the flood relay's cost.

Remark 11 (Coordinated first visits: measured, not yet proved)**.** The delay term $4F' \min(K, N_\mathrm{b})$ charges the early pulls of each bin, and E37 confirms what they are in practice: far-apart agents pull the same bin while they still see it as empty ("blind" pulls: 412 with a server, 543 on the ring, 581 on the path). A 1-scalar "I am exploring bin j" claim cannot help the published-only relay. With $\gamma < 1$, a pull into a bin seen as empty is published in the same step, and a claim would travel no faster. A rule that needs *no* messages can help. Each bin has an owner fixed by a shared hash, and an agent leaves a bin it sees as empty to its owner until it has had that bin available A times. With $A = 10$, the extra regret over the server is -41 to -12 on all five graphs of E36 (it was 118–643), blind pulls fall to the server's level, and the relay sends 1.6× the scalars. The rule also helps a server (by 112). Against a server with the same rule, the relay's extra regret is 71–100, nearly flat in F'. What a proof needs: the rule uses counts and contexts only, so Lemmas 24 and 25 hold unchanged. Lemma 14 fails for a *held* pull (the best bin may be held), so held pulls need their own term, at most A per agent and bin. Bounding that term by something smaller than $NN_\mathrm{b}A$ is open.

6 Communication

We count *scalars*: integers or reals sent over one edge in one direction. This is the measure used by Wang et al. (2020).

Lemma 28 (Number of epochs)**.** *For each bin j, let A_j be the set of epochs at which some agent's alarm condition (1) holds for bin j. Then*

$$|A_j| \;\le\; 1 + \log_{1+\gamma}(NT), \qquad |\mathcal{E}| \;\le\; 1 + N_\mathrm{b}\big(1 + \log_{1+\gamma}(NT)\big).$$

Proof. List $A_j = \{a_1 < a_2 < \dots\}$ and let $g_p = G_j(a_p)$.

Every alarm epoch has $g_p \ge 1$. An alarm for bin j at a_p means some $\delta_{l,j}(a_p) > 0$, so at least one pull into j happened by round a_p.

Growth between consecutive alarm epochs. Let $a_{p+1} = \beta_m$. The alarm at β_m is by some agent l, with $\delta_{l,j}(\beta_m) > \gamma \max(G_j(\epsilon_m), 1)$. The pulls counted in $\delta_{l,j}(\beta_m)$ happen in rounds $(\epsilon_m, \beta_m]$. So

$$g_{p+1} = G_j(\beta_m) \;\ge\; G_j(\epsilon_m) + \delta_{l,j}(\beta_m) \;>\; (1+\gamma)\, G_j(\epsilon_m),$$

using $G_j(\epsilon_m) \ge g_p \ge 1$. That inequality holds because $a_p \le \beta_{m-1}$ and $a_p \in \mathcal{E}$, so $a_p \le \epsilon_m$. Hence $g_{p+1} > (1+\gamma)\, g_p$.

Counting. So $NT \ge g_p > (1+\gamma)^{p-1} g_1 \ge (1+\gamma)^{p-1}$, which gives $p < 1 + \log_{1+\gamma}(NT)$. Finally, every epoch other than β_0 belongs to some A_j, and only bins that are actually pulled can trigger. □

Theorem 29 (Communication)**.** *The total number of scalars sent by Dec-ZoomSIB-Relay is at most*

$$\underbrace{2|E|\, N\,(d+1)}_{\textit{Phase 1}} \;+\; \underbrace{2|E|\,|\mathcal{E}|\,\big(1 + N(3N_\mathrm{b}+1)\big)}_{\textit{Phase 2}} \;=\; O\big(|E|\, N\, N_\mathrm{b}^2 \log_{1+\gamma}(NT) + |E|\, N\, d\big).$$

For a fixed grid (fixed N_b) and fixed γ it grows only logarithmically in T. By contrast, any protocol in which every agent sends its N_b-bin table to its neighbours every round sends $\Theta(|E|\, N_\mathrm{b}\, T)$ scalars.

Proof. Idea. Under flooding every message crosses each edge at most once, so it suffices to count messages: N records in Phase 1, then one alarm and N snapshots per epoch. Lemma 28 bounds the number of epochs.

Under flooding, each message is forwarded by each agent at most once, to each neighbour. So one message crosses each of the $2|E|$ directed arcs at most once.

Phase 1. There are N records, each with $d+1$ scalars.

Phase 2. For each epoch, every agent sends at most one alarm over each arc: either its own alarm or the first one it receives, never both. This is one scalar per arc. It also generates N snapshots. Each snapshot holds at most N_{b} changed bins, at 3 scalars each (index, count increment, sum increment), plus one label. Multiply by $2|E|$ and by the number of epochs from Lemma 28. □

Remark 12. Snapshots can instead be summed up a BFS spanning tree and broadcast back down. Each tree edge then carries at most $3N_{\mathrm{b}}+1$ scalars per direction per epoch, which removes the factor $N|E|/(N-1)$: $O\big((NN_{\mathrm{b}}+|E|)\,N_{\mathrm{b}}\log_{1+\gamma}(NT)\big)$. We keep flooding in the main statement because it is what the robustness results of Section 7 need. The bound counts every pulled bin as changed in every snapshot, so it is loose. In practice a snapshot contains only the bins the agent pulled since the last epoch.

Remark 13 (Communication at the rate-optimal parameters)**.** Corollary 18 takes $\Delta=(NT)^{-1/3}$ and $\gamma=\gamma_0/N$. Then $N_{\mathrm{b}}=\tilde{O}((NT)^{1/3})$ and $\log_{1+\gamma}(NT)\le(1+\gamma_0)N\log(NT)/\gamma_0$, so Theorem 29 gives $\tilde{O}\big(|E|\,N^2(NT)^{2/3}\big)$ scalars. This is polynomial in T, but still below the $\Theta(|E|\,N_{\mathrm{b}}T)=\tilde{\Theta}(|E|(NT)^{1/3}T)$ of per-round sharing once $T\gtrsim N^{7/2}$. The factor N in the number of epochs is a trade-off, not a slack in the proof. A constant γ needs N times fewer epochs, but costs $\sqrt{1+N\gamma}=\Theta(\sqrt{N})$ in the UCB term of Theorem 17. Each agent's snapshots also hold at most T bin entries in total, one per pull, so the snapshot traffic is never more than $O(|E|\,NT)$ whatever γ is.

6.1 Memory and identities

The flood relay in the code keeps a copy of every agent's table: NN_{b} counts and sums per agent. It also needs agent IDs to avoid double counting, and every agent sees every other agent's statistics. The epoch protocol needs much less.

Proposition 30 (Bounded memory)**.** *(i) In the epoch protocol, each agent can store, instead of any per-agent table: the summed table $G(e), G^S(e)$ of its reference epoch; the running sums of the at most two epochs whose snapshots are still arriving; its own counts at those epochs; and the labels (epoch, sender) of the snapshots received for the epochs in flight. That is $O(N_{\mathrm{b}}+N)$ numbers, against $O(NN_{\mathrm{b}})$ for the flood relay.*

(ii) If $\mathcal{G}$ is static, the snapshots can instead be summed up a BFS spanning tree and broadcast back down. Then each agent stores $O(N_{\mathrm{b}})$ numbers plus its tree neighbours. No global IDs are needed, and an agent only ever sees partial sums from its subtree, never another agent's own table. With slot length $L=\Lambda+2h$ (h the tree depth, $h\le D$), Theorem 17 holds with 4Λ replaced by $2L$ in $c_{N,\Lambda}$. The communication becomes that of the remark after Theorem 29.

Proof. (i) An agent acts on $G(e)+[O_i(t-1)-O_i(e)]$ (Section 3), so it needs only the sum over agents, not the terms. It can add each snapshot to the running sum of its epoch on arrival. The labels prevent adding the same snapshot twice under flooding. They can be dropped once the epoch is complete, which by Lemma 9 happens within L rounds. So at most two epochs are in flight at any time.

(ii) The alarm still takes Λ rounds to reach everyone. The partial sums then take h rounds to reach the root and h rounds to come back down, so every agent has $G(e)$ within $\Lambda + 2h = L$ rounds of the epoch. That is all Lemma 9 uses. Lemmas 15 and 16 only use the slot length, with $2L$ in place of 4Λ. A node stores its own snapshot and the sum from its children. □

7 Robustness

The proofs of Sections 4 and 6 use the graph only through Assumption 2. So any communication model with a known delivery bound Λ gets Theorems 17 and 29 with that Λ. It suffices to compute Λ in each case.

Lemma 31 (Delivery time under weaker connectivity)**.** *Assumption 2 holds with the following Λ:*

(i) Directed graphs. *If $\mathcal{G}$ is a static, strongly connected directed graph with directed diameter D, then $\Lambda = D$.*

(ii) Time-varying graphs, union connectivity. *Let the edge set E_s change with the step s. Suppose that for every window of B consecutive steps the union graph $\bigcup_{s \in window} E_s$ is connected. Use* persistent flooding*: every agent re-sends each message it holds to its current neighbours in every step, until Λ steps after the message was created. Then $\Lambda = B(N-1)$.*

(iii) Time-varying graphs, stable subgraph. *Suppose instead that every window of B consecutive steps contains a connected spanning subgraph H that is present at every step of the window, with diameter $D_B \le B$. Then plain flooding gives $\Lambda = D_B$.*

(iv) Link delays. *Let $\mathcal{G}$ have diameter D, and suppose every message sent over a link is delivered within τ steps. Then $\Lambda = \tau D$.*

Proof. (i) This is the proof of Lemma 1, with "neighbour" replaced by "in-neighbour" and distance by directed distance.

(ii) Let the message be created in step s, and cut the steps from s on into consecutive windows of B steps. Let U_r be the set of agents that hold the message after r windows, with $U_0 = \{a\}$. If $U_r \neq V$, the union graph of window $r+1$ is connected, so it has an edge $\{u, v\}$ with $u \in U_r$ and $v \notin U_r$. Under persistent flooding, u sends the message across that edge at the step where it appears, so $v \in U_{r+1}$. Each window adds at least one agent, so $U_{N-1} = V$, which happens by step $s + B(N-1) - 1$.

(iii) Apply the proof of Lemma 1 to H in the window that starts at step s. Every forwarding it uses happens by step $s + D_B - 1 \le s + B - 1$, so inside the window, where H is present.

(iv) The induction of Lemma 1, with each hop taking at most τ steps. □

Remark 14 (The diameter of the union graphs does not help)**.** Under union connectivity alone, Λ cannot be bounded by the diameter of the union graphs. In window r, let the only edges form a star centred at an agent c_r that does not yet hold the message, and let them all appear at the last step of the window. Every window's union graph is a star, of diameter 2. But in each window only c_r is newly informed: the informed leaves reach c_r in that last step, and c_r cannot forward within the same step. So delivery takes $B(N-1)$ steps, and the bound in (ii) is tight. A short simulation confirms this for several (N, B).

Corollary 32 (Robustness)**.** *Under any model of Lemma 31, with slot length $L = 2\Lambda$ for the corresponding Λ, Theorem 17 and Corollary 18 hold verbatim. In particular, a time-varying graph costs an additive $\tilde{O}\big(N^2 B\,(NT)^{1/3}\big)$ in regret under union connectivity (case (ii)), and $\tilde{O}\big(N D_B (NT)^{1/3}\big)$ with a stable subgraph (case (iii)). Records arrive late but are never wrong, so the leading $(NT)^{2/3}$ term does not change. Theorem 29 holds verbatim in cases (i) and (iv), and in case (iii) with*

$|E|$ *replaced by* $\max_s |E_s|$. *In case (ii) it holds with an extra factor* Λ *and with* $|E|$ *replaced by* $\max_s |E_s|$, *because each message is re-sent for* Λ *steps.*

Proof. Lemma 9 is the only place where delivery is used, and it uses only Assumption 2. Everything after it depends only on the epoch structure. For communication, plain flooding sends each message at most once from each agent to its neighbours at that step, and persistent flooding sends it over each present arc at most once per step, for Λ steps. □

Remark 15. Random link failures, such as each link failing independently with probability q per step (one of the settings of E18), give a random delivery time. With persistent flooding, the corollary then applies on the event that every message used is delivered within a chosen Λ. A union bound over messages and over the hops of a shortest path shows that this event has probability at least $1 - 1/(NT)$ when $\Lambda \gtrsim (D + \log(NT))/(1 - q)$. The complement adds a term of order L_f to the bound.

8 Agreement

So far every agent froze the same direction $\hat{\theta}_0$. Approximate Phase-1 methods such as consensus or push-sum leave each agent k with its own direction $\hat{\theta}_k$. Bins are then computed with different projections, but all tables are still added up bin by bin.

Theorem 33 (Cost of disagreement)**.** *Run Phase 2 of Dec-ZoomSIB-Relay with agent* k *binning by its own* $\hat{\theta}_k$, $\|\hat{\theta}_k\|_1 = 1$, *on the same grid. Let* $\eta = \max_k \|\hat{\theta}_k - \theta^*\|_1$, *and let* $\bar{\eta} \geq 0$ *satisfy* $\mathbb{P}(\eta > \bar{\eta}) \leq \delta$. *Then with probability at least* $1 - 3\delta$, *the bound of Theorem 17 holds with the projection term replaced by*

$$4L_{f'} X_{\max} \bar{\eta}\, NT.$$

In particular, let $\hat{\theta}$ *be the exact pooled direction and suppose* $\max_k \|\hat{\theta}_k - \hat{\theta}\|_1 \leq \eta_{\text{agree}}$. *Then* $\bar{\eta} = \varepsilon_0(NT_0) + \eta_{\text{agree}}$ *is valid, and the extra regret relative to exact pooling is at most* $4L_{f'} X_{\max} \eta_{\text{agree}}\, NT$. *It is zero when Phase 1 pools exactly.*

Proof. Idea. The directions enter the regret proof only through the projection error ε_{p}.

Lemmas 11, 13, 14 and 16 were stated for arbitrary directions $\hat{\theta}_k$. The directions enter only through $\varepsilon_{\text{p}} = X_{\max} \max_k \|\hat{\theta}_k - \theta^*\|_1$:

- Lemma 11(ii) puts the true index of a bin-j pull by *any* agent in the same widened interval $\tilde{I}_j$. So pooling the bin-j statistics of different agents is still pooling pulls whose means lie in $[m_j - \omega, m_j]$.
- Lemma 13 concerns only noise, which does not depend on the directions.
- Lemma 16 concerns only counts.

The proof of Theorem 17 then goes through on $E_x \cap E_\eta \cap \{\eta \leq \bar{\eta}\}$, which has probability at least $1 - 3\delta$, with $\varepsilon_{\text{p}} \leq X_{\max}\bar{\eta}$. For the second statement, the triangle inequality $\|\hat{\theta}_k - \theta^*\|_1 \leq \|\hat{\theta} - \theta^*\|_1 + \|\hat{\theta}_k - \hat{\theta}\|_1$ gives $\eta \leq \varepsilon_0(NT_0) + \eta_{\text{agree}}$ on E_θ, so $\mathbb{P}(\eta > \bar{\eta}) \leq \delta$. Compare with the exact-pooling bound. □

Remark 16 (One flooding pass makes Phase 1 exact)**.** Any approximate Phase-1 method, such as consensus or push-sum, can be followed by one flooding pass in which every agent floods its own *local* Stein sum b_k. After Λ more rounds every agent holds all N local sums (Assumption 2) and computes the exact pooled direction, so Lemma 10 and Theorem 33 apply with $\eta_{\text{agree}} = 0$. The cost is Λ extra rounds of uniform play, at most $2L_f N\Lambda$ regret, and $2|E|N(d+1)$ scalars. Since flooding the local sums alone already gives the exact direction, an approximate method helps only if agents must start Phase 2 before the flood completes.

8.1 Stopping Phase 1 without a server

DBG stop Phase 1 adaptively: they watch the drift of $\hat{\theta}$ at checkpoints and stop once it is small. On a graph, letting each agent apply this rule to its own estimate fails (E28b). Agents stop at different times, with different directions. The fix is to make every agent apply the rule to the *same* numbers.

Proposition 34 (Serverless stopping with agreement)**.** *Let ρ be a stopping rule for the server protocol. It is checked at fixed checkpoint rounds $c_1 < c_2 < \dots$, and its decision at c_m is a deterministic function of the pooled Stein sums at $c_1, \dots, c_m$. The serverless version works as follows:*

(a) at each checkpoint c_m, every agent floods its local sum $b_k(c_m)$ and its sample count;
(b) at round $c_m + \Lambda$, every agent holds all N sums, forms the pooled sums, and applies ρ;
(c) if ρ stops at c_m, every agent freezes $\hat{\theta}(c_m) = \sum_k b_k(c_m) / \| \sum_k b_k(c_m) \|_1$ and starts Phase 2 at round $c_m + \Lambda + 1$.

Then all agents stop at the same checkpoint as the server would, and freeze the same direction, so $\eta_{\text{agree}} = 0$. Compared with the server run of ρ, the regret grows by at most $2L_f N\Lambda$, and each checkpoint costs $2|E|N(d+1)$ scalars.

Proof. By Assumption 2, every sum flooded at c_m has reached every agent by round $c_m + \Lambda$. Every agent therefore computes the same pooled sums, which are those the server would compute (Lemma 10). Since ρ is deterministic, every agent makes the server's decision. The only difference from the server run is that Phase 1 lasts Λ more rounds, during which agents keep pulling uniformly. That costs at most $2L_f N\Lambda$ on E_x, and the extra pulls are not used. Communication is as in Phase 1 of Theorem 29. □

Remark 17 (What the proposition does and does not give)**.** It removes the agreement problem completely: no "any" or "all" rule is needed, and no agent needs to detect that the others are ready. It also inherits every guarantee of the server rule. If the server rule always stops at a checkpoint where NT_0 satisfies the sample-size condition of Lemma 2, Theorem 17 holds with T_0 the stopping checkpoint and $T_0 + \Lambda$ in the Phase-1 term, after a union bound over the checkpoints. With geometric checkpoints there are $O(\log NT)$ of them. Whether adaptive stopping beats a fixed T_0 is a separate question, and it is a server-side one. In E33 ($N = 16$; torus, ring, path) the serverless rule stopped exactly D rounds after the server's rule in every paired trial, with zero disagreement. But both adaptive rules lost to a fixed $T_0 = 60$: by $\times 1.29$–1.39 (server) and $\times 1.41$–1.49 (serverless), at tolerances 0.15 and 0.05. The drift rule itself is the weak point, not the graph.

Remark 18 (Link to E28)**.** Theorem 33 is linear in the disagreement and has no threshold. E28 (fig28, panel D) shows regret rising steeply once disagreement exceeds about 10^{-2}. That is consistent, since the bound only becomes comparable to the other terms once $X_{\max}\eta_{\text{agree}}NT$ is of order $(NT)^{2/3}$. E28b confirms Remark 16: one flooding pass removes the whole penalty of one-step consensus (from $\times 1.25$ to $\times 1.08$ on the ring, and from $\times 1.34$ to $\times 1.09$ on the path).

9 Privacy of Phase 1

In Phase 1 each agent sends a single message, its Stein sum b_k, and its actions do not depend on its data (uniform play). That makes Phase 1 easy to privatise: one dose of Gaussian noise per agent suffices. We protect each Phase-1 sample (x, y) in the usual sense of $(\varepsilon_p, \delta_p)$-differential privacy (DP), with neighbouring datasets differing in one sample of one agent.

Theorem 35 (Private Phase 1)**.** *Let each agent flood* $\tilde{b}_k = b_k + Z_k$ *instead of* b_k*, where* $Z_k \sim \mathcal{N}(0, \sigma_p^2 I_d)$ *and* $\sigma_p = 2\tau\sqrt{d}\,\sqrt{2\log(1.25/\delta_p)}/\varepsilon_p$*, with* $\varepsilon_p \le 1$*. All agents use* $\hat{\theta}_0 = \sum_k \tilde{b}_k / \|\sum_k \tilde{b}_k\|_1$*. Then:*

(i) everything the network ever sends and does, in both phases, is $(\varepsilon_p, \delta_p)$*-DP with respect to each Phase-1 sample;*

(ii) with probability at least $1 - 4\delta$*, Theorem 17 holds with* $\varepsilon_0(NT_0)$ *replaced by*

$$\varepsilon_0(NT_0)\,(1+\kappa), \qquad \kappa = \frac{\sqrt{6\,d\,N\log(1.25/\delta_p)}}{\varepsilon_p},$$

provided the sample-size condition of Lemma 2 holds with the extra factor,

$$(1+\kappa)\,C_\theta d\sqrt{\log(2d/\delta)/n} \le \mu^*/2, \qquad n = NT_0.$$

With T_0 *re-balanced as in Corollary 18, the leading term becomes* $\tilde{O}\big(((1+\kappa)\,d\,NT)^{2/3}\big)$*.*

Proof. (i) The truncation φ_τ keeps every coordinate of every summand in $[-\tau, \tau]$. Changing one sample therefore changes b_k by at most 2τ per coordinate, so by at most $2\tau\sqrt{d}$ in ℓ_2. The Gaussian mechanism with the stated σ_p makes $\tilde{b}_k$ $(\varepsilon_p, \delta_p)$-DP. Each sample enters exactly one b_k. Everything else is post-processing of the $\tilde{b}_k$ and of data independent of the Phase-1 sample: Phase-1 actions are uniform, Phase 2 uses Phase 1 only through $\hat{\theta}_0$, and the Phase-2 data are new.

(ii) Write $n = NT_0$. The noise in the pooled mean is $\bar{Z} = \sum_k Z_k/n$, with coordinates $\mathcal{N}(0, N\sigma_p^2/n^2)$. With probability $\ge 1-\delta$, $\|\bar{Z}\|_1 \le d\sqrt{N}\sigma_p\sqrt{2\log(2d/\delta)}/n$. Substituting τ from Lemma 2 gives

$$\|\bar{Z}\|_1 \;\le\; \frac{4\sqrt{3}\,d^{3/2}\sqrt{NM(\sigma^2 + L_f^2)\log(1.25/\delta_p)}}{\varepsilon_p\sqrt{n}}.$$

DBG's Lemma 2.1 (the normalisation step inside Lemma 2) turns an unnormalised error $\epsilon \le \mu^*/2$ into $4\epsilon/\mu^*$. Here ϵ is the Stein error plus $\|\bar{Z}\|_1$. Since $C_\theta \ge 2\sqrt{2}\sqrt{M(\sigma^2 + L_f^2)}$, the ratio of $4\|\bar{Z}\|_1/\mu^*$ to $\varepsilon_0(n)$ is at most κ. The rest of the proof of Theorem 17 uses $\hat{\theta}_0$ only through this error. Re-balancing T_0 multiplies the projection constant by $1+\kappa$, which gives the last claim. □

Remark 19 (The price, and what is left)**.** The factor $\kappa \propto \sqrt{dN}/\varepsilon_p$ is the price of *local* privacy: without a trusted aggregator, each of the N messages carries its own noise, and the noises add up. Privacy therefore costs a factor $(dN)^{1/3}/\varepsilon_p^{2/3}$ in the leading term, not a constant. Phase 2 is harder. Agents act on their own raw rewards, so the counts in their snapshots, and even their alarms, leak information about those rewards. Protecting Phase-2 rewards would require agents to act only on privatised statistics, for example the tree-based mechanism that Dubey & Pentland (2020) use for linear bandits. We leave that open.

Remark 20 (The constants matter: E35)**.** The noise is calibrated to the truncation level τ, which is set from the worst-case L_f on $|z| \le X_{\max}$. For the quadratic link on $|z| \le 6$, $L_f = 48$, while typical summands are of order one. The bound ε_0 contains the same loose constant, so κ (a ratio of two bounds) understates the practical cost. In E35 ($N = 16$, $d = 10$, ring), $\varepsilon_p \le 32$ leaves the direction essentially random. At $\varepsilon_p = 128$ ($\kappa \approx 0.8$), regret is $\times 12$ at $T_0 = 60$ and $\times 5$ at $T_0 = 960$; the longer Phase 1 does buy back part of the loss, as the theorem says. A practical private Phase 1 needs a much smaller clipping level, trading bias for noise. We leave this open.

10 Summary

Result	Statement	Evidence
Thm 5	$\Omega((NT)^{2/3})$ for any graph and protocol	E22
Thm 7, Cor 8	ring/path: also $\Omega(N \min(T, D)^{1/3})$	–
Thm 17, 19, 22	epochs: $\tilde{O}((dNT)^{2/3} + N\Lambda K)$, w.h.p.	E31, E36
Thm 27	published-only relay, any γ: $\tilde{O}((dNT)^{2/3} + \sqrt{1+N\gamma}(NT)^{2/3} + KF')$	E23, E36
Thm 29, Prop 30	$O(\lvert E\rvert N N_{\mathrm{b}}^2 \log_{1+\gamma} NT)$ scalars, $O(N_{\mathrm{b}} + N)$ memory	E15, E20, E31
Cor 32	directed, delayed, time-varying graphs: replace D by Λ	E18, E32
Thm 33, Prop 34	disagreement costs $\le 4L_{f'}X_{\max}\eta NT$; stopping with agreement	E28, E33
Thm 35	private Phase 1, direction error $\times(1+\kappa)$	E35

Open. (a) Upper and lower graph terms do not meet: the upper bounds pay an additive, horizon-free $O(KF')$ (linear in the farness), while the light-cone lower bound grows like $N \min(T, D)^{1/3}$ and only bites when $T \lesssim \sqrt{ND}$. (b) The dependence on K: the bounds are linear in K (from $\sum_j q_j \le K$), while E36 measures growth like $K^{0.4}$–$K^{0.5}$. (c) Privacy of Phase-2 rewards, and a practical clipping level for private Phase 1.